

AIM TRAINER

```
import math

import random

import time

import pygame

pygame.init()

WIDTH , HEIGHT = 800 , 600

WIN = pygame.display.set_mode((WIDTH , HEIGHT))

pygame.display.set_caption("AIM TRAINER")

TARGET_INCREMENT = 400

TARGET_EVENT = pygame.USEREVENT

TARGET_PADDING = 30

BG_COLOR = (0 , 25 ,40)

LIVES = 5

TOP_BAR_HEIGHT = 50

LABEL_FONT = pygame.font.SysFont("comicsans" , 24)

class Target:

    MAX_SIZE = 30

    GROWTH_RATE = 0.2

    COLOR = "red"

    SECOND_COLOR = "WHITE"

    # constructor

    def __init__(self , x ,y):

        self.x = x

        self.y = y

        self.size = 0

        self.grow = True

    def update(self):

        if self.size + self.GROWTH_RATE >= self.MAX_SIZE:

            self.grow = False
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if self.grow:

    self.size += self.GROWTH_RATE

else:

    self.size -= self.GROWTH_RATE


def draw(self , win):

    pygame.draw.circle(win , self.COLOR , (self.x , self.y) , self.size)

    pygame.draw.circle(win , self.SECOND_COLOR , (self.x , self.y) , self.size * 0.8)

    pygame.draw.circle(win , self.COLOR , (self.x , self.y) , self.size * 0.6)

    pygame.draw.circle(win , self.SECOND_COLOR , (self.x , self.y) , self.size * 0.4)


def collide(self , x ,y):

    dis = math.sqrt((x -self.x )**2 + (y - self.y )**2)

    return dis <= self.size


def draw(win ,targets):

    win.fill(BG_COLOR)


    for target in targets:

        target.draw(win)


def format_times(secs):

    milli = math.floor(int(secs * 1000 % 1000 ) /100)

    seconds = int(round(secs % 60 , 1))

    minutes = int(secs // 60)


    return f"{minutes:02d}:{seconds:02d}:{milli:02d}"


def draw_top_bar(win , elapsed_time , targets_pressed , misses):

    pygame.draw.rect(win , "grey" , (0,0 ,WIDTH ,TOP_BAR_HEIGHT ))

    time_label = LABEL_FONT.render(f"Time : {format_times(elapsed_time)}" , 1 , "black")

    speed = round(targets_pressed / elapsed_time , 1)

    speed_label = LABEL_FONT.render(f"Speed : {speed} t/s" , 1 , "black")

    hits_label = LABEL_FONT.render(f"Hits : {targets_pressed} " , 1 , "black")

    lives_label = LABEL_FONT.render(f"lives : {LIVES - misses} " , 1 , "black")

```

```
win.blit(time_label , (5 ,5 ))
```

```
win.blit(speed_label , (250 ,5 ))
```

```
win.blit(hits_label , (450 ,5 ))
```

```
win.blit(lives_label , (600 ,5 ))
```

```
def end_screen(win , elapsed_time , targets_pressed , clicks):
```

```
    WIN.fill(BG_COLOR)
```

```
    time_label = LABEL_FONT.render(f"Time : {format_times(elapsed_time)}" , 1 , "white")
```

```
    speed = round(targets_pressed / elapsed_time , 1)
```

```
    speed_label = LABEL_FONT.render(f"Speed : {speed} t/s" , 1 , "white")
```

```
    hits_label = LABEL_FONT.render(f"Hits : {targets_pressed} " , 1 , "white")
```

```
    accuracy = round(targets_pressed / clicks * 100 , 1)
```

```
    accuracy_label = LABEL_FONT.render(f"Accuracy: {accuracy}% " , 1 , "white")
```

```
win.blit(time_label , (get_middle(time_label) ,100))
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```
win.blit(speed_label , (get_middle(speed_label),200))
```

```
win.blit(hits_label , (get_middle(hits_label),300 ))
```

```
win.blit(accuracy_label , (get_middle(accuracy_label),400))
```

```
pygame.display.update()
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```
run = True
```

```
while run:
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```
    for event in pygame.event.get():
```

```
        if event.type == pygame.QUIT:
```

```
            quit()
```

```
        if event.type == pygame.KEYDOWN:
```

```
            run = False
```

```
            break
```

```
def get_middle(surface):
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```
    return WIDTH/ 2- surface.get_width()/2
```

```
def main():
```

```
    run = True
```

```

targets = []

clock = pygame.time.Clock()

target_pressed = 0

clicks = 0

misses = 0

start_time = time.time()

pygame.time.set_timer(TARGET_EVENT , TARGET_INCREMENT)

while run:

    clock.tick(60)

    click = False

    mouse_pos = pygame.mouse.get_pos()

    elapsed_time = time.time() - start_time

    for event in pygame.event.get():

        if event.type == pygame.QUIT:

            run = False

            break

        if event.type == TARGET_EVENT:

            x = random.randint(TARGET_PADDING , WIDTH-TARGET_PADDING)

            y = random.randint(TARGET_PADDING + TOP_BAR_HEIGHT , HEIGHT-TARGET_PADDING)

            target = Target(x,y)

            targets.append(target)

        if event.type == pygame.MOUSEBUTTONDOWN:

            click = True

            clicks +=1

    for target in targets:

        target.update()

        if target.size <= 0:

            targets.remove(target)

            misses +=1

# ekk ekk coordinate lega voh

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    if click and target.collide(*mouse_pos):

        targets.remove(target)

        target_pressed +=1

if misses >= LIVES:

    # end the game

    end_screen(WIN , elapsed_time , target_pressed , clicks)


draw(WIN , targets)

draw_top_bar(WIN , elapsed_time , target_pressed ,misses)

pygame.display.update()

pygame.quit()

if __name__ == "__main__":

    main()
```